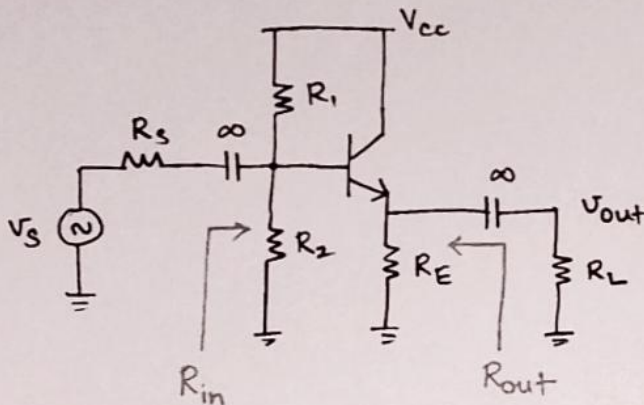


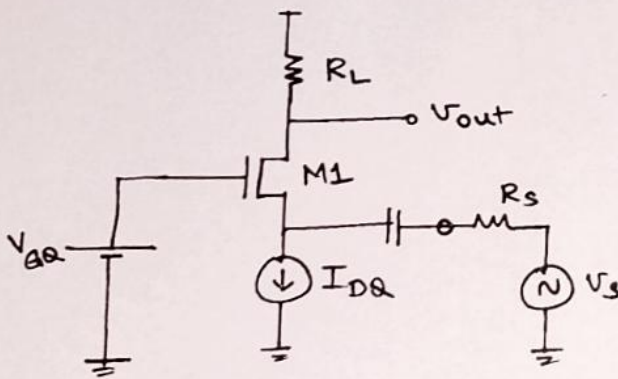
- ① Calculate the input resistance, output resistance and voltage gain of the common collector amplifier shown below.

(Your answer should be in terms of $R_s, R_1, R_2, R_E, R_L, r_\pi, g_m$ etc.)



$$A_v = \frac{v_{out}}{v_{in}}$$

- ② Consider the common-gate amplifier below :- (Assume $\lambda = 0$ for M1)



- Calculate the voltage gain v_{out}/v_s .
- Calculate the power gain P_{out}/P_s .
- Find the value of g_m which maximizes the power gain.

- ③ Construct the y -parameter two-port matrix for the CG amplifier of problem-2.